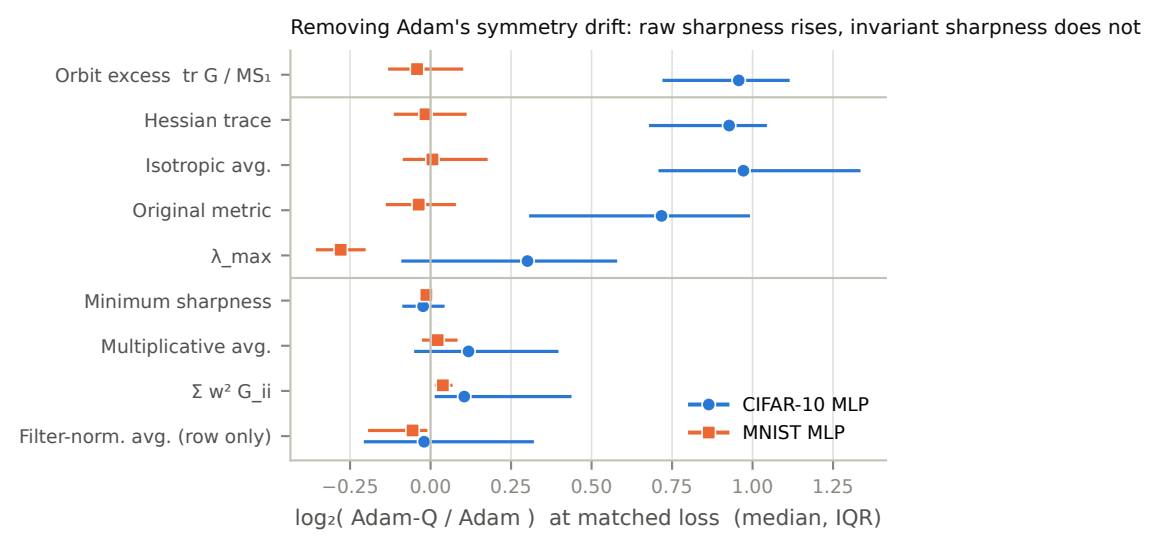
Removing Adam's symmetry drift: raw sharpness rises, invariant sharpness does not
Orbit excess tr G / MS₁
Hessian trace
Isotropic avg.
Original metric
λ_max
Minimum sharpness
Multiplicative avg.
Σ w² G_ii
Filter-norm. avg. (row only)
CIFAR-10 MLP
MNIST MLP
−0.25
0.00
0.25
0.50
0.75
1.00
1.25
log₂(Adam-Q / Adam) at matched loss (median, IQR)